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Original Article

autism

Autism
2021, Vol. 25(5) 1246–1261
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DOI: 10.1177/1362361320984896
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Effects of autism acceptance training on explicit and implicit biases toward autism

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Abstract

Non-autistic adults often hold explicit and implicit biases toward autism that contribute to personal and professional challenges for autistic people. Although previous research indicates that non-autistic adults with higher autism knowledge and familiarity express more inclusionary attitudes, it remains unclear whether training programs designed to promote autism acceptance and understanding affect subsequent implicit and explicit biases toward autism. In this study, non-autistic adults ($N = 238$) completed an autism acceptance training featuring factual information and engaging first-person narratives, a general mental health training not mentioning autism, or a no-training control, then responded to surveys assessing their autism knowledge, stigma, and impressions of autistic adults, and completed a novel implicit association task about autism. Non-autistic adults in the autism acceptance training condition reported more positive impressions of autistic adults, demonstrated fewer misconceptions and lower stigma about autism, endorsed higher expectations of autistic abilities, and expressed greater social interest in hypothetical and real autistic people. However, training had no effect on implicit biases, with non-autistic adults associating autism-related labels with unpleasant personal attributes regardless of training condition. These findings suggest that the autism acceptance training program in this study, designed to increase autism knowledge and familiarity among non-autistic people, holds promise for reducing explicit but not implicit biases toward autism.

Lay abstract

Autistic adults face prejudice from non-autistic people. They are often judged unfairly and left out of social activities because of their differences. This can make it difficult for autistic people to make friends and find jobs. Some training programs have tried to teach autistic people to act more like non-autistic people to help them gain acceptance. Fewer have focused on teaching non-autistic people how to be more autism friendly. In this study, we used a short training video that teaches people about autism. The video was created with the help of autistic adults and included clips of real autistic people. We found that non-autistic people who watched this video had better knowledge about autism and showed more autism-friendly attitudes than those who watched a video about mental health or those who did not watch any video. They were more open to having a relationship with an autistic person and had more positive beliefs about autism. However, our video did not affect people's unconscious attitudes about autism. People in our study connected autism with unpleasant traits, even if they had watched the autism training video. This suggests that teaching non-autistic people about autism may promote more autism-friendly attitudes, but some beliefs may be harder to change.

Keywords

first impressions, inclusion, intervention, stigma

Introduction

Although attitudes about autism may be improving (D. White et al., 2019), stigma remains common (Butler & Gillis, 2011; Gillespie-Lynch et al., 2020; Hinshaw & Stier, 2008). Stigma occurs when differences among groups are stereotyped and labeled as unfavorable, leading to social exclusion and discrimination for group members (Link & Phelan, 2001). This process occurs rapidly for non-autistic (NA) people when socially evaluating their autistic peers (Sasson et al., 2017), with NA adults making almost instantaneous unfavorable inferences about the

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attributes of autistic people based on their social, behavioral, and communicative differences (Faso et al., 2015; Grossman, 2015; Sasson et al., 2017; Stagg et al., 2014).

Such biases among NA adults are associated with a reluctance to interact with autistic people (Cage & Burton, 2019; Morrison et al., 2019, 2020; Sasson et al., 2017), creating social barriers to inclusion that can be detrimental to their personal and professional well-being (Cage et al., 2018, 2019; Kapp et al., 2011). For instance, autistic adults commonly experience difficulties establishing and maintaining relationships despite expressing strong desires for friendships and romantic relationships (Berns, 2017; Eaves & Ho, 2008), and these unfulfilled social needs can contribute to the high levels of loneliness, depression, and anxiety they often experience (Hedley et al., 2018; Mazurek, 2014; Sumiya et al., 2018). Professionally, autistic adults face many challenges securing employment commensurate with their skills and abilities (Taylor et al., 2015). They are frequently evaluated less positively by potential employers (Baldwin et al., 2014; Scott et al., 2017), employed at lower rates than those with other disabilities (Shattuck et al., 2012), and paid lower hourly wages (Cimera & Cowan, 2009). This extends to autistic adults with little to no cognitive impairment, who are rarely paid at the same wage as NA peers (Taylor & Seltzer, 2011). Taken together, these findings suggest that social structures (e.g. systemic barriers and lack of accommodation), not just individual characteristics of autistic people, contribute to poor personal and professional outcomes in autism (Woods, 2017).

Because attitudes about autism are highly variable among NA people (Morrison et al., 2019), recent work has begun to identify factors associated with more inclusionary attitudes, with the hope that these may serve as mechanisms for developing more accommodating social environments (Gillespie-Lynch et al., 2020; Sasson & Morrison, 2019). A great deal of recent attention has focused on whether diagnostic disclosure increases acceptance of autistic differences among NA people (for a review, see Thompson-Hodgetts et al., 2020). However, disclosing one's diagnosis is a fraught decision for many autistic adolescents and adults (Lindsay et al., 2019; O'Connor et al., 2019; R. White et al., 2020), who fear that autism-related stigma can harm how they are viewed by others (Jones et al., 2015) and increase their risk for experiencing discrimination and victimization (Zeedyk et al., 2014). Indeed, although NA perceptions of autistic people do improve on average when informed of their diagnosis (Maras et al., 2019; Sasson & Morrison, 2019), disclosure is not always a beneficial strategy. In some contexts, disclosure can have little to no effect (R. White et al., 2020) or even be detrimental depending on the autism-related stigma held by the person receiving the disclosure (Morrison et al., 2019). Emphasizing disclosure as a mechanism for improving attitudes among NA people also shifts the onus to the autistic person to be treated equally and respectfully and absolves the NA individual from working toward greater acceptance and accommodation.

Two characteristics linked to more inclusionary attitudes among NA adults are autism familiarity and knowledge (Gillespie-Lynch et al., 2015; Obeid et al., 2015). Familiarity, defined as experiencing high quantity and quality of personal contact with people with clinical conditions, is associated with reduced stigma toward those conditions (Corrigan & Nieweglowski, 2019), including autism (Gardiner & Iarocci, 2014), though familiarity may not always extend to attitudes about inclusion of autistic people (Bottema-Beutel et al., 2019). For people with the highest familiarity of autism, autistic individuals themselves, negative trait assessments of autistic people do not reduce their desire to interact with and befriend them as they do for NA adults (DeBrabander et al., 2019), suggesting that biases about autistic self-presentation do not invariably translate to exclusionary behaviors. In addition, NA adults with more autism knowledge, defined as having a high factual understanding of autism and a low tendency to endorse common autism misconceptions, exhibit lower levels of stigma toward autistic people (Morrison et al., 2019) and evaluate them more favorably (Sasson & Morrison, 2019). Increasing autism knowledge is also associated with reductions in autism-related stigma (Obeid et al., 2015). However, NA people are often poor at judging their level of autism knowledge, with those who know the least overestimating their knowledge the most (McMahon et al., 2020), suggesting that some NA people may fail to recognize their own misconceptions about autism that could affect their perceptions of, and interactions with, autistic people.

A great deal of work in other fields has examined whether increasing familiarity and knowledge can reduce stigma toward marginalized populations (Pettigrew & Tropp, 2008), with training programs targeting biases for race (Gonzalez et al., 2017), gender (Jackson et al., 2014), and sexual orientation (Dasgupta & Rivera, 2008) all demonstrating some efficacy among adults. The findings of these studies suggest that discriminatory attitudes can be reduced through short-term exposure to personal vignettes or stories about individuals in a stigmatized group. Although previous research has demonstrated that online training can improve autism knowledge and decrease stigma among college students (Gillespie-Lynch et al., 2015), there remains a need to determine whether training programs can reduce explicit and implicit biases in evaluations of actual autistic people, as opposed to evaluations of diagnostic labels or vignette characters. Importantly, explicit and implicit biases may differentially influence the expression of stigma and discriminatory behaviors. While explicit biases are controlled and influenced by social desirability (Dovidio et al., 1997; Stier & Hinshaw, 2007), implicit biases are automatic and more likely to

impact unconscious or impulsive decisions (Strack & Deutsch, 2004). Furthermore, previous research has found that measures of explicit biases toward autism may underreport stigma (Dickter, Burk, Zeman, et al., 2020; Hinshaw & Stier, 2008), with significant implicit biases persisting among raters who reported no explicit biases. These findings suggest that NA individuals may maintain negative attitudes toward autistic people even if they are not consciously aware of them. As a result, it is possible that increasing autism knowledge and familiarity among NA people may reduce explicit biases about autism to a greater degree than implicit ones. While educational programs designed to increase accurate understanding of autism among NA people often fail to produce significant changes in implicit biases (Bast et al., 2020; Dickter, Burk, Anthony, et al., 2020), a recent study found greater reductions in implicit biases for parents of NA children compared to parents of autistic children (Dickter, Burk, Anthony, et al., 2020). This finding raises the possibility that educational training programs to reduce implicit biases toward autism may be especially effective for individuals who are less familiar with autism.

This study examines whether NA college students completing a brief autism acceptance training module express lower explicit and implicit biases toward autism than NA college students in two separate control conditions. Participants were assigned to one of three conditions (autism acceptance training, a more general mental health training that did not mention autism, or a no-training control), with participants in the two training conditions viewing a 25-min video about either autism or general mental health that included factual information and engaging first-person narratives. Participants in all three conditions then viewed videos of autistic adults and evaluated their first impressions of them, responded to questionnaires assessing their autism knowledge and stigma, and completed a novel implicit association task measuring implicit attitudes about autism. We predicted that participants who completed autism acceptance training would demonstrate greater autism knowledge, report less stigma toward autism, form more positive first impressions of autistic people, and indicate greater social interest in interacting with them, compared to participants in the general mental health training condition and control condition. Furthermore, we predicted that autism acceptance training would reduce implicit biases about autism relative to both the mental health and no-training control conditions, but these effects would be smaller relative to its effect on explicit biases given that implicit biases are more resistant to change and often require more protracted intervention to diminish (Devine et al., 2012). Support for these findings would provide potential for a short, easily accessible tool to decrease both explicit and implicit biases toward autistic people within the general population.

Method

Participants

A total of 281 university undergraduates were recruited for participation. Twenty-six participants were excluded due to a software crash unrelated to the study tasks, which was later resolved through a firmware update and 17 were excluded for indicating on a questionnaire that they did not pay close attention to the training video or give their best effort completing the outcome measures, resulting in a final sample of 238 undergraduates. Participants ranged in age from 18 to 57 years ($M = 21.50$, $SD = 4.67$) and had estimated IQs in the average to above-average range ($M = 108.59$, $SD = 9.55$) as approximated by the Reading Subtest of the Wide Range Achievement Test-3 (WRAT-3; Wilkinson, 1993). Participants predominantly identified as White (40%) or Asian (39%), with the remaining participants identifying as Black (8%), American Indian (1%), or an unspecified "Other" race (12%). Most participants identified as non-Hispanic (83%) and female (69%) with the rest identifying as male (30%), and non-binary or gender fluid (1%). In total, 72% of participants reported having at least one parent with a college degree or greater.

No included participant self-reported a diagnosis of autism or a developmental disability in themselves or in a first-degree relative. All participants provided written informed consent and received course credit for participating. The study was approved by the university institutional review board (IRB).

Procedure

Participants were first administered the reading subtest of the WRAT-3 (Wilkinson, 1993), which correlates highly with IQ (Wiens et al., 1993). Next, participants were randomly assigned to one of the three conditions: (1) autism acceptance training ($n = 77$), (2) a mental health training ($n = 77$), or (3) a no-training control condition ($n = 84$). Participants in the two training conditions watched a 25-min-long narrated video relating to either autism or general mental health (described below), while individuals in the control condition proceeded immediately to the next steps of the study. The three training groups did not differ significantly on participant age ($p = 0.414$) or estimated IQ ($p = 0.697$), or in reported gender, race, ethnicity, or maternal or paternal education using a chi-square test of independence (gender: $p = 0.449$; race: $p = 0.193$; ethnicity: $p = 0.930$; maternal education: $p = 0.353$; paternal education: $p = 0.275$).

Participants next viewed a set of 20 videos of autistic adults recruited from a local service provider (17 males, 3 females; mean age = 24.5 years, mean IQ on the WASI = 106.4) participating in the High-Risk Social Challenge Task (Gibson et al., 2010), a 60-s unrehearsed mock audition for a reality TV show. Each video was trimmed to

focus on the first 10 s of substantive behavior and speech, which served as a representative "thin slice" for each autistic adult (Ambady & Rosenthal, 1992). More information about the autistic adults appearing in the videos, as well as details about the creation and validation of the videos themselves, have been previously reported (Morrison et al., 2019; Sasson et al., 2017; Sasson & Morrison, 2019).

Participants viewed each video one at a time in a random order. To assess whether potential benefits of training would differ depending on whether an individual's diagnosis was disclosed, half of the videos were accompanied by an accurate diagnostic label ("this person has autism") and the other half were accompanied by no label. A counterbalanced version was also created that reversed which videos were labeled versus unlabeled, and participants were randomly assigned to one of the two versions. After viewing each video, participants completed the First Impressions Scale (Sasson et al., 2017) measuring their trait assessments of, and social interest in, the autistic adult in each video.

Following the video ratings, participants completed measures assessing their autism knowledge, their stigma toward autism, their perceptions of functional abilities in autistic individuals, and a task assessing implicit biases toward autism. These measures were completed in a random order and were followed by a demographics questionnaire.

Training conditions

Autism acceptance training. Originally developed by Dr Grace Iarocci's Autism & Developmental Disorders Lab at Simon Fraser University with the input and feedback of an autistic adult featured in the module and then edited, shortened, and improved for sensitivity through collaboration with an autism self-advocacy group, this training was designed to facilitate autism acceptance in high school students and consists of a 38-slide PowerPoint presentation that introduces participants to some of the difficulties that autistic individuals often face (Boucher et al., 2020). The presentation was modified in this study to be more relevant to a college-aged audience, and narration was added to improve engagement. The video covers a range of topics, including diagnostic characteristics of autism, sensory sensitivities, autistic strengths, difficulties autistic people may experience in college, neurodiversity, and ways to increase inclusion and acceptance of autistic people. Four engaging clips featuring first-person accounts are presented throughout the video, providing insight into the different ways that autistic people experience the world and the challenges they face. These clips include a young autistic woman discussing her communication differences, a NA father interacting with his autistic son, and a young autistic man explaining the benefit of his stimming behaviors and the pressure he feels to suppress them in public. The video is approximately 25 min long.

Mental health training. Participants in the mental health training condition viewed a 25-min-long video created as part of the Centerville Cares suicide prevention program (Community Film Project, 2018). This video presents factual information on attention-deficit hyperactivity disorder, depression, suicide, self-harm, post-traumatic stress disorder, and addiction, as well as first-person accounts of five individuals sharing their experiences with these conditions. Autism is never mentioned. This video was selected to control for the possibility that sensitivity training toward clinical conditions generally, rather than autism specifically, affects stigma and biases toward autistic individuals.

Measures

Explicit biases and knowledge about autism. *The First Impressions Scale* (Sasson et al., 2017) is a 10-item questionnaire assessing the rater's initial impressions of a target individual. Six items assess traits: awkwardness, attractiveness, trustworthiness/honesty, dominance, likeability, and intelligence. Four items assess the rater's behavioral intentions and interest in interacting with the target individual (e.g. "I would hang out with this person in my free time"). Participants responded on a 4-point Likert-type scale, with higher scores corresponding to more favorable impressions. Recent studies using this scale have consistently found less favorable impressions of autistic adults relative to NA controls (DeBrabander et al., 2019; Morrison et al., 2019; Sasson et al., 2017). The scale served as a measure of explicit biases about autistic adults and had acceptable internal consistency in this study ($\alpha = 0.76$).

The Autism Knowledge Scale (Gillespie-Lynch et al., 2015) is a 12-item questionnaire assessing an individual's factual knowledge of autism. Participants read statements reflecting common facts (e.g. "Autism is more frequently diagnosed in males than females") or misconceptions (e.g. "With the proper treatment, most children diagnosed with autism can eventually outgrow the disorder") about autism and rated their agreement or disagreement with each statement on a 5-point scale from −2 to 2. Scores on each item were summed to create a composite score, with higher scores denoting a more accurate understanding of autism. Item-level comparisons were also conducted, consistent with prior work (Gillespie-Lynch et al., 2015). This scale's internal consistency ($\alpha = 0.66$; Gillespie-Lynch et al., 2017) is lower than the 0.7 threshold commonly used to designate "good" internal reliability and performed similarly in this study ($\alpha = 0.69$).

The Social Distance Scale (SDS; Gillespie-Lynch et al., 2015) is a six-item questionnaire assessing stigma toward autism. Participants were asked to rate their likelihood of socially engaging with an autistic individual across various scenarios (e.g. "How willing would you be to marry or date a person with autism?") on a 4-point Likert-type

Table 1. Block order for autism IAT.

Block order	Counterbalanced order	Type of judgment	Left side	Right side
1	4	Diagnostic discrimination	Typically developing	Autism spectrum
2	2	Personality trait discrimination	Pleasant	Unpleasant
3	5	Prejudice consistent combination	Typically developing or pleasant	Autism spectrum or unpleasant
4	1	Diagnostic discrimination, reversed order	Autism spectrum	Typically developing
5	3	Prejudice inconsistent combination	Autism spectrum or pleasant	Typically developing or unpleasant

IAT: Implicit Association Test.

scale. Scores on each item were summed to create a total score, with higher scores reflecting greater stigma toward autism. In line with previous work, comparisons on the SDS were also conducted at the item level (Gillespie-Lynch et al., 2015). Internal consistency of this scale is strong ($\alpha = 0.83$–0.87; Gillespie-Lynch et al., 2015, 2017) and was also high in this study ($\alpha = 0.87$).

The *Specific Levels of Functioning Scale* (SLOF; Schneider & Struening, 1983) assesses adult adaptive functioning abilities. Here, we included 30 items across four subscales: Interpersonal Relationships, Social Acceptability, Activities, and Work Skills. Within the Social Acceptability subscale, one item regarding repetitive behaviors was excluded. Participants selected the response best describing their perceptions of how typical an item would be for an autistic person (e.g. "communicates effectively" or "has employable skills"). Scores were derived by summing item scores within each subscale, with lower scores indicating greater perceptions of disability. The SLOF has been used extensively in studies of autistic individuals or other clinical populations (Harvey et al., 2011; Leifker et al., 2011) and showed high internal consistency in this study ($\alpha = 0.90$).

Implicit biases about autism. Implicit Association Tests (IAT; Greenwald et al., 1998) are used to probe automatic associations between cognitive concepts (e.g. Black, White) and attributes (e.g. good, bad). In this study, a novel IAT was developed and administered using PsyToolkit (Stoet, 2010, 2017) to examine whether participants unconsciously associate autism diagnostic labels with unpleasant personal attributes and to determine whether autism acceptance training decreases these associations. The target words used for each category are displayed in Table 2. Diagnostic terms associated with the autism spectrum (autism, autistic, Asperger's) were chosen to align with common and preferred terms of autistic people and their families (Kenny et al., 2016). Although Asperger's Syndrome was removed from *Diagnostic and Statistical Manual of Mental Disorders* (5th ed.; *DSM*-5; American Psychiatric Association, 2013), this term was included based on its wide adoption by autistic people and their families (Kenny et al., 2016), as well as evidence that it is not perceived differently by NA people than other autism labels (Brosnan & Mills, 2016; Morrison et al., 2019). Personal attribute terms were drawn from previous literature regarding common stigma-related attributions made toward autism and other clinical conditions. These fell into three categories of "unpleasant" attributes: concerns about safety (dangerous, antisocial, unpredictable; Corrigan et al., 2006; Link & Phelan, 2001); dependency (needy, helpless; Huws & Jones, 2010); and atypicality (awkward, weird, creepy; Grossman, 2015; Sasson et al., 2017). Furthermore, qualitative pilot data determined that these and related terms were volunteered by undergraduates at our university when discussing various clinical conditions, providing additional external validity for the selection of these attributes. Antonyms for these terms were then selected for the pleasant attributes.

Table 2. IAT stimulus categories and words.

Category	Stimulus words
Diagnosis	
Autism spectrum	Autistic, Autism, Asperger's
Typically developing	Normal, Neurotypical, No diagnosis
Personality traits	
Pleasant	Nice, friendly, likeable, safe, popular, honest, compassionate, independent
Unpleasant	Awkward, creepy, weird, dangerous, antisocial, needy, unpredictable, helpless

IAT: Implicit Association Test.

Our IAT task used a counterbalanced five-block design (see Table 1), and stimulus words for each concept and attribute are presented in Table 2. In the first block, participants categorized words (e.g. "Neurotypical," "Asperger's") based on diagnostic concepts ("Typically Developing" and "Autism Spectrum") presented on the left and right sides of the screen. Participants pressed the "e" key to categorize a

term as "Typically Developing" or the "i" key to categorize it as "Autism Spectrum." The second block focused on personal attributes, with participants categorizing words (e.g. "friendly," "awkward") as either "Pleasant" or "Unpleasant." In the third block, the concept and attribute categories were displayed simultaneously in a prejudice consistent manner (e.g. "Typically Developing or Pleasant" on the left of the screen, "Autism Spectrum or Unpleasant" on the right of the screen), and participants categorized both concept and attribute words one at a time as they appeared. The fourth block was identical to the first block, except the positioning of the two concepts was reversed, with participants pressing the "e" key to categorize a word as "Autism Spectrum" and the "i" key to categorize a word as "Typically Developing." In the fifth block, the diagnostic concept and attribute terms were displayed simultaneously in a prejudice inconsistent manner (e.g. "Autism Spectrum or Pleasant" on the left, "Typically Developing or Unpleasant" on the right), and again participants categorized stimulus words into their respective categories.

Within blocks, each stimulus word was displayed on the center of the screen in a random order over 20 trials, with each word appearing once before any repeats occurred. The inter-trial interval was 100 ms. Scoring for the IAT is based on the D-score, representing the standardized difference in response times for prejudice inconsistent and prejudice consistent pairings (Greenwald et al., 2003); positive scores indicate a relationship between a concept and negative attributes. The IAT showed high internal consistency for all five blocks (block 1: $\alpha = 0.93$, block 2: $\alpha = 0.94$, block 3: $\alpha = 0.94$, block 4: $\alpha = 0.87$, block 5: $\alpha = 0.94$). The slightly lower, but still high, internal consistency in block 4 was primarily driven by greater response latencies for the first few trials, which likely reflects participants' adjustment to the reversal of category orientations (left vs right side) for this block.

Analysis plan

First impressions ratings. Multi-level modeling (MLM) was used with restricted maximum likelihood (REML) estimation in order to assess whether autism acceptance training was associated with more favorable first impressions of autistic adults relative to mental health training and the no-training control. In this model, we utilized a cross-classified random effects model, as each NA participant rated multiple autistic adults, making our data crossed and non-independent. Effects were estimated for training condition (i.e. autism acceptance training, mental health training, or control), stimulus participant label (i.e. accurate vs no label), and the interaction between these factors, yielding *F* tests for both main effects and the interaction. Significant effects for condition were followed up using Bonferroni-corrected pairwise comparisons. Estimates of effect size were computed by regressing standardized outcomes on dummy-coded predictors, with the autism acceptance condition as the comparison group.

Knowledge and stigma measures. One-way analyses of variance (ANOVAs) were used to determine whether autism knowledge and stigma differed as a function of training condition. Significant results were followed with post hoc Tukey tests for pairwise comparisons between the control, autism acceptance training, and mental health training conditions.

IAT. Raw response latencies were obtained for each IAT block and inspected to determine if any trials or participants exceeded common guidelines for inclusion (Greenwald et al., 2003). None did. Next, D-scores were calculated and analyzed using the revised scoring method outlined in Greenwald et al. (2003). To obtain D-scores, participants' mean response latencies for the prejudice consistent block (Autism Spectrum + unpleasant) were subtracted from those of the prejudice inconsistent block (Autism Spectrum + pleasant). These difference scores were then divided by the standard deviation of participants' response latencies across the two blocks, resulting in a standardized score that accounts for individual variability in latencies. One-sample *t*-tests were used to determine whether D-scores differed significantly from 0, indicating a significant association between autism labels and negative or unpleasant attributes. To determine whether the strength of associations differed significantly across training conditions, a one-way ANOVA was used, with D-scores as the dependent variable and training condition as the factor.

Associations between explicit and implicit biases. Pearson correlations were used to determine the association between implicit and explicit measures of stigma.

Community involvement. The first author conceived of the study, led data collection, conducted the analysis, and drafted the manuscript. She identifies as neurodivergent and, along with the second author, worked closely with a local autism service provider to establish relationships with autistic community members and recruit them for research studies. The second and last author identify as neurotypical. They contributed to the design and interpretation of the study and helped compose the manuscript. A research team of undergraduate and master's students, including four students who identify as autistic, helped with data collection for the study and contributed to discussions that formulated the project and interpreted the findings. Finally, an autistic adult contributed to the autism acceptance training module, which was then further improved through consultation with an autistic-led autism advocacy group.

Table 3. Fixed effects of training condition, diagnostic label, and interactions on first impressions ratings.

	Intercept		Main effect of training condition		Main effect of diagnostic label		Interaction	
	F	*p*	*F*	*p*	*F*	*p*	*F*	*p*
Awkward	758.79	<**0.001**	2.77	0.07	23.95	<**0.001**	0.17	0.85
Attractive	1414.67	<**0.001**	4.19	**0.02**	56.65	<**0.001**	0.33	0.72
Dominant	1076.86	<**0.001**	2.19	0.11	2.90	0.09	1.10	0.33
Likeable	3873.80	<**0.001**	2.72	0.07	60.63	<**0.001**	1.34	0.26
Intelligent	5380.99	<**0.001**	9.52	<**0.001**	24.63	<**0.001**	1.23	0.29
Trustworthy	6234.52	<**0.001**	2.33	0.10	35.67	<**0.001**	0.38	0.68
Live near	6307.93	<**0.001**	1.63	0.20[a]	28.95	<**0.001**	0.93	0.40
Hang out	2219.13	<**0.001**	5.18	**0.01**	30.83	<**0.001**	0.18	0.84
Sit near	6346.16	<**0.001**	3.57	**0.03**	15.74	<**0.001**	1.66	0.19
Conversation	3039.66	<**0.001**	4.68	**0.01**	39.03	<**0.001**	0.60	0.55

Note. Bolding indicates significance at $p < .05$.
[a]Removal of ten influential outliers resulted in a change in statistical significance ($F(225, 2) = 3.17$, $p = 0.044$).

Results

Preliminary analysis

Scale scores were evaluated for skew and kurtosis prior to analysis, with acceptable values ranging from −2 to 2. Scores for the Autism Knowledge Scale, the SDS, and all four SLOF subscales, as well as IAT D-scores, fell within acceptable limits for skewness (range: −0.32 to 1.05) and kurtosis (range: −0.52 to 1.70). Homogeneity of variances across the three training conditions was assessed using Levene's test, and variances did not differ significantly for scores on the Autism Knowledge Scale, SDS, the four SLOF subscales, or the IAT D-scores ($ps > 0.09$).

For First Impressions Scale ratings, normality of the residuals was evaluated for each item based on skewness and kurtosis. Skewness fell within acceptable limits for all 10 items (range: −1.12 to 0.49), and kurtosis was acceptable for 6 of 10 items (range: 0.30 to 1.73). The remaining four items showed moderate to high kurtosis (trust: 2.92, smart: 2.26, live near: 6.03, sit next to: 4.29), suggesting the presence of outliers. Using the HLMdiag package in RStudio (R Core Team, 2012), influential observations for each item were identified, and analyses were run with these data omitted. One previously non-significant predictor reached significance when outliers were removed (see Table 3 footnote), but all other patterns of findings were preserved, suggesting that substantive conclusions were largely unimpacted by kurtosis.

First impressions ratings

Fixed effects for the impacts of training condition, diagnostic label, and their interaction on first impressions ratings are reported in Table 3. On the six trait items, participants in the autism acceptance training condition rated autistic individuals as significantly more attractive ($p = 0.016$, $\beta = -0.30$) and more intelligent ($p < 0.001$, $\beta = -0.47$) compared to raters in the mental health training condition (Figure 1). Intelligence ratings also were significantly higher in the no-training condition than the mental health training condition ($p = 0.042$, $\beta = -0.18$). No significant effects of training were found for perceptions of likeability, trustworthiness, dominance, or awkwardness. For the four "behavioral intentions" items, participants in the autism acceptance training reported a stronger desire to hang out with autistic adults than participants in both the mental health training ($p = 0.021$, $\beta = -0.29$) and the no-training condition ($p = 0.013$, $\beta = -0.30$), and a stronger desire to start a conversation with autistic individuals compared to participants in the mental health training ($p = 0.038$, $\beta = -0.28$) and the no-training conditions ($p = 0.017$, $\beta = -0.29$). There was also a significant main effect of training on participants' comfort sitting near autistic adults, but this did not survive post hoc corrections. Participant impressions of autistic adults were significantly more favorable when informed of their diagnosis across all three training conditions (see Table 3). This occurred for all four behavioral intent items and for every trait item except for dominance. However, no significant interactions between diagnostic label and training condition were found.

Explicit beliefs about autism

Scores on the Autism Knowledge Scale differed significantly post-training across the three groups ($F(2, 220) = 5.80$, $p = 0.003$), with participants in the autism acceptance training condition demonstrating more accurate understanding of autism compared to those in the general mental health training condition ($p = 0.003$, $d = 0.59$) but not the no-training control ($p = 0.073$, $d = 0.35$; $M_{ASD} = 13.83$, $M_{MH} = 11.08$, $M_{Control} = 12.06$). At the item level (Figure 2), participants completing the autism acceptance training were less likely than those in the other two conditions to endorse the misconceptions that autistic people are

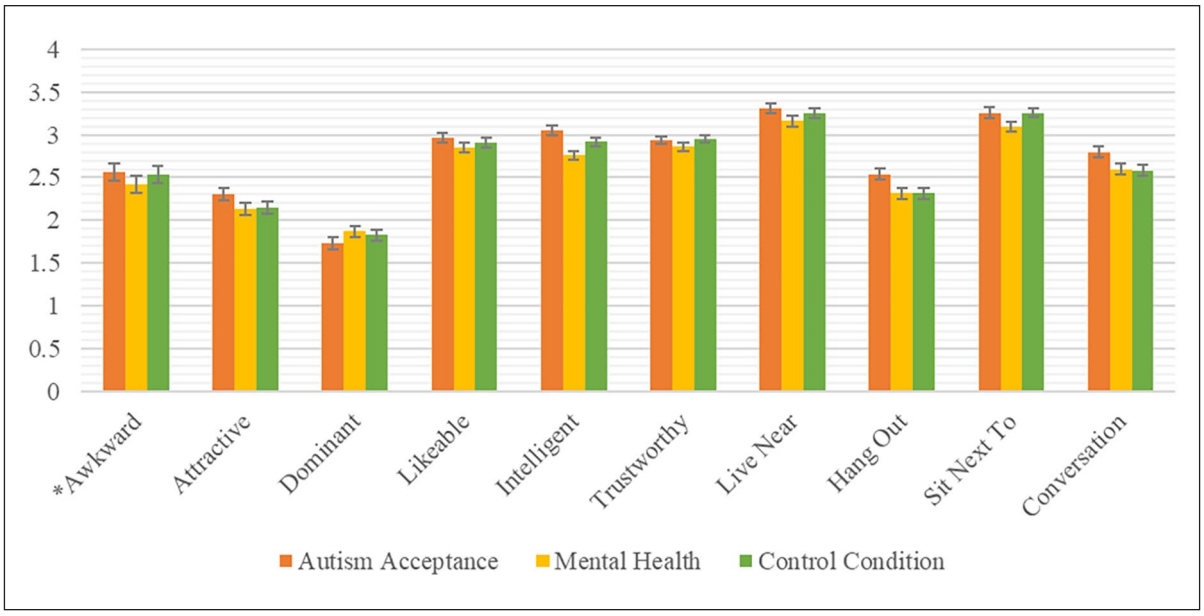


Figure 1. First impressions ratings by intervention group.
*Values for awkwardness were reverse scored. Therefore, higher scores on each item reflect more positive first impressions. Error bars represent standard error.

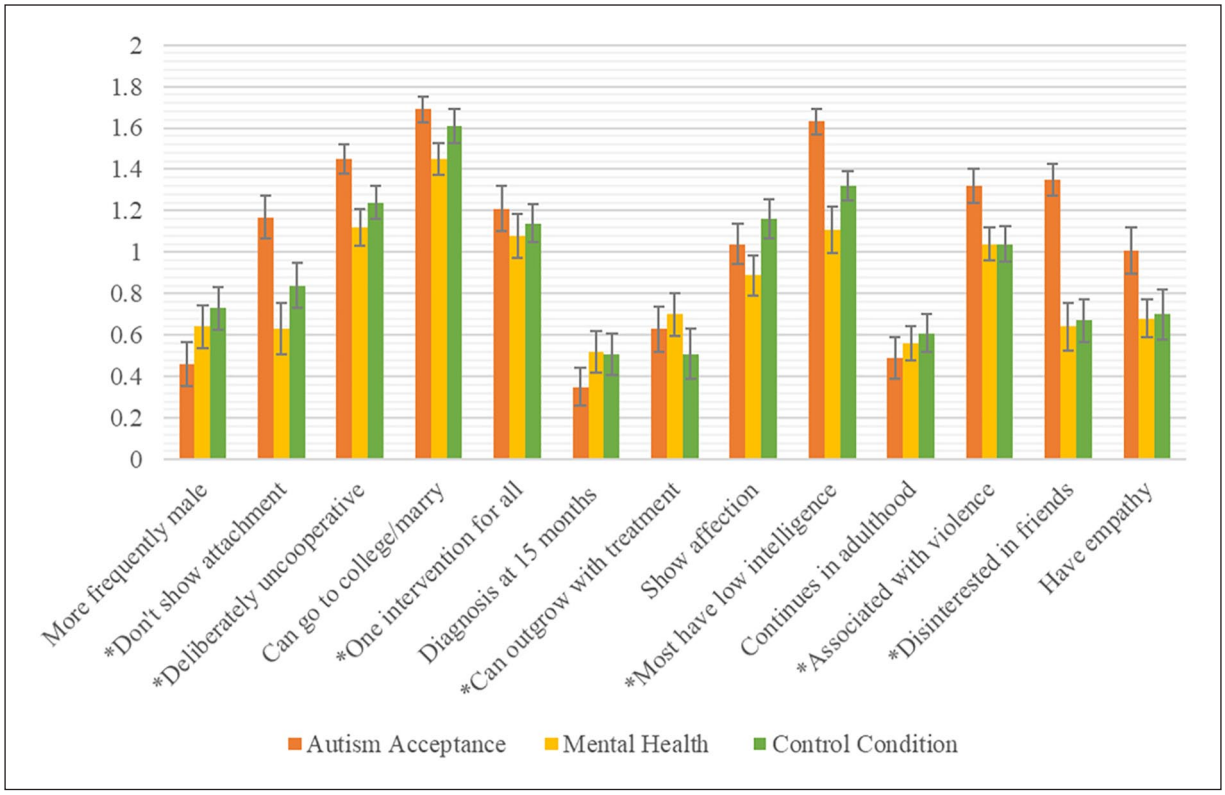


Figure 2. Autism knowledge ratings by intervention group.
Items derived from the Autism Knowledge Scale (Gillespie-Lynch et al., 2015).
*Signifies reverse-scored items, with higher values corresponding to greater disagreement with the statement. For all other items, higher scores reflect greater agreement with the statement. Error bars represent standard error.

not interested in friendships (Control: $p < 0.001$, $d = 0.85$; MH: $p < 0.001$, $d = 0.85$) or have low intelligence (Control: $p = 0.024$, $d = 0.52$; MH: $p < 0.001$, $d = 0.67$), and were more likely to disagree that autistic people are deliberately uncooperative ($p = 0.014$, $d = 0.48$) and do not show attachment behaviors ($p = 0.003$, $d = 0.56$) compared to participants in mental health training, and less likely than participants in the no-training control to believe that autistic people are violent ($p = 0.043$, $d = 0.38$). In contrast, scores did not differ significantly for any item between the mental health training and no-training conditions.

Summed scores on the SDS also differed significantly across training conditions ($F(2, 207) = 5.30$, $p = 0.006$). Participants in the autism acceptance training had lower stigma about interacting with autistic individuals ($M = 8.79$) compared to those in mental health training ($M = 10.54$; $p = 0.007$; $d = 0.53$) and the no-training control conditions ($M = 10.18$; $p = 0.036$; $d = 0.45$). They also were more accepting of marrying or dating an autistic person ($M = 1.99$) compared to the mental health training ($M = 2.39$; $p = 0.022$; $d = 0.45$) and no-training conditions ($M = 2.38$; $p = 0.020$; $d = 0.47$), and were more willing to befriend an autistic person ($M_{ASD} = 1.24$, $M_{MH} = 1.50$, $M_{Control} = 1.42$; $p = 0.038$; $d = 0.43$), have an autistic person marry into the family ($M_{ASD} = 1.40$, $M_{MH} = 1.82$, $M_{Control} = 1.70$; $p = 0.008$; $d = 0.54$), and spend an evening socializing with an autistic person ($M_{ASD} = 1.27$, $M_{MH} = 1.64$, $M_{Control} = 1.51$; $p = 0.001$; $d = 0.61$) compared to the mental health training participants. The mental health training and no-training groups did not differ significantly on any items.

Scores on the SLOF differed significantly across training conditions for the Social Acceptability ($M_{ASD} = 22.86$, $M_{MH} = 20.74$, $M_{Control} = 21.17$; $p = 0.003$), Activities ($M_{ASD} = 39.62$, $M_{MH} = 36.90$, $M_{Control} = 38.41$; $p = 0.017$), and Work Skills ($M_{ASD} = 23.11$, $M_{MH} = 20.83$, $M_{Control} = 21.65$; $p = 0.003$) subscales but not the Interpersonal Relationships subscale ($M_{ASD} = 21.96$, $M_{MH} = 20.90$, $M_{Control} = 20.88$; $p = 0.245$). Participants in the autism acceptance training believed that autistic people engage in more socially acceptable behaviors compared to those in the mental health training and the no-training groups (MH: $p = 0.005$, $d = 0.54$; control: $p = 0.025$, $d = 0.31$) and believed that autistic individuals can participate in more activities ($p = 0.013$, $d = 0.52$) and possess more work abilities ($p = 0.002$, $d = 0.58$) than participants in the mental health training group.

Implicit biases about autism

An average IAT D-Score of 0.28 was found across the three groups. This effect differed significantly from zero ($t(236) = 8.22$, $p < 0.001$, $d = 0.53$), indicating that participants responded more quickly when autism labels were paired with unpleasant attributes compared to when these attributes were paired with labels associated with typical development. However, the magnitude of D-scores did not differ significantly between training conditions ($M_{ASD} = 0.21$, $M_{MH} = 0.33$, $M_{Control} = 0.29$; $p = 0.34$). Mean response latencies for each intervention condition are displayed in Figure 3.

Associations between explicit and implicit biases

Correlations between first impressions ratings and each explicit and implicit measure are reported in Table 4. In general, more positive first impressions of autistic people were associated with greater autism knowledge, lower autism stigma, and higher perceptions of autistic functional abilities. Higher implicit biases toward autism were associated with higher ratings of awkwardness and lower ratings of attractiveness for autistic people but did not correlate significantly with other first impression items.

As can be seen in Table 5, the explicit measures of autism knowledge, autism stigma, and perceptions of functional abilities correlated moderately well with one another, but implicit biases toward autism did not significantly correlate with any of the explicit measures except for two items on the first impression scale.

Discussion

This study evaluated the effects of a brief autism acceptance training module on explicit and implicit biases toward autistic people among NA adults. Relative to a general mental health training module and a no-training control, participants completing autism acceptance training demonstrated fewer misconceptions and lower stigma about autism, higher expectations of autistic functional abilities, and more positive first impressions of autistic adults. These effects for explicit biases did not extend to implicit biases. NA participants, regardless of training condition, implicitly associated autism-related labels with unpleasant personal attributes. Taken together, these findings suggest that training programs designed to increase autism knowledge and understanding demonstrate promise for reducing explicit but not implicit biases toward autism.

Participants completing autism acceptance training expressed more inclusive and accepting attitudes about autism relative to those in the mental health training and no-training conditions. They were less likely to endorse misconceptions that autistic people are prone to violence, have low intelligence, and do not desire friendships, and were more likely to consider autistic differences as socially acceptable compared to the two other conditions. They also believed that autistic people could engage in a higher number of activities and have better work skills compared to participants in the mental health training condition. Finally,

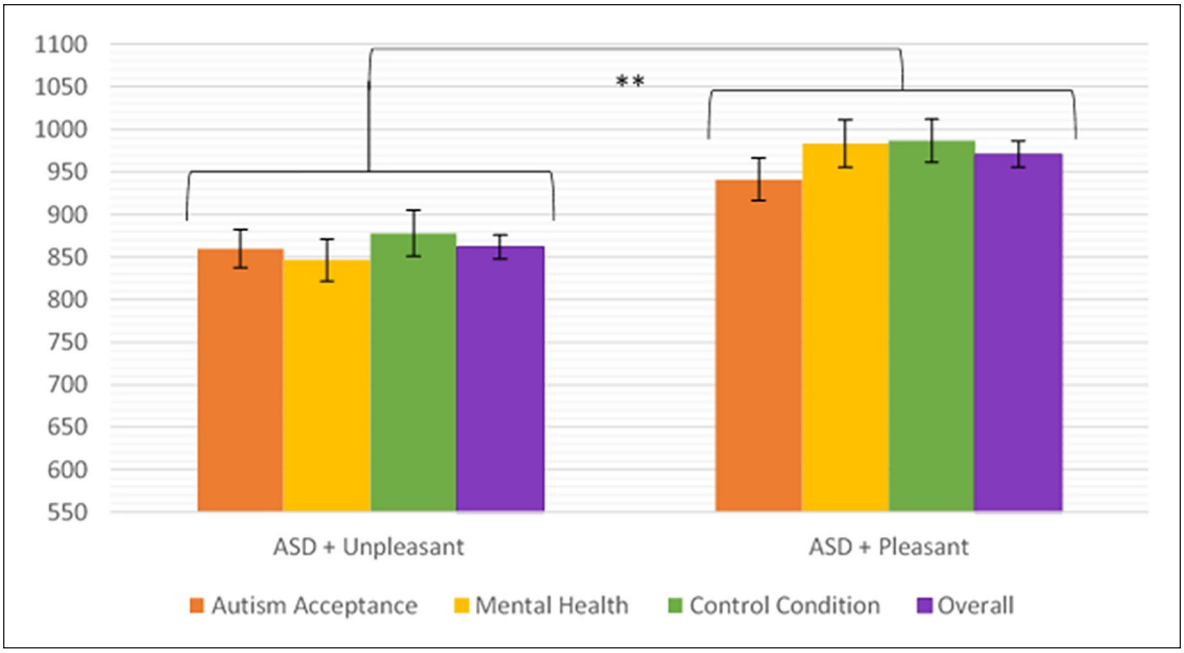


Figure 3. Response latencies for the ASD Implicit Association Test.
Image represents latencies (in milliseconds) for the prejudice consistent (ASD + unpleasant) and prejudice inconsistent (ASD + pleasant) conditions.
**$p < 0.001$.

Table 4. Correlations between first impressions ratings and measures of autism stigma.

	Autism knowledge[a]	Autism stigma[b]	SLOF Interpersonal	SLOF Acceptability	SLOF Activities	SLOF Work	IAT D-score
Awkward	0.00	0.20**	−0.19**	−0.20**	−0.14*	−0.20**	0.19**
Attractive	0.09	0.20**	0.21**	0.12	0.10	0.19**	−0.17*
Trustworthy	0.29**	−0.34**	0.29**	0.15*	0.31**	0.34**	−0.06
Dominant	−0.18**	0.11	−0.12	−0.26**	−0.25**	−0.29**	0.01
Likeable	0.28**	−0.46**	0.38**	0.21**	0.38**	0.39**	−0.01
Smart	0.23**	−0.36**	0.23**	0.18**	0.26**	0.35**	−0.01
Live near	0.34**	−0.42**	0.15*	0.18**	0.33**	0.25**	0.04
Hang out	0.19**	−0.43**	0.27**	0.20**	0.24**	0.33**	−0.11
Sit next to	0.30**	−0.39**	0.21**	0.18**	0.32**	0.28**	0.02
Conversation	0.09	−0.29**	0.24	0.13	0.21**	0.24**	−0.06

SLOF: Specific Levels of Functioning Scale; IAT: Implicit Association Test.
[a]Scores derived from the Autism Knowledge Scale (Gillespie-Lynch et al., 2015).
[b]Scores derived from the Social Distance Scale (Gillespie-Lynch et al., 2015).
*$p < 0.05$.
**$p < 0.01$.

NA adults completing autism acceptance training expressed greater interest in interacting with both hypothetical and real autistic people on several metrics, including openness to romantic relationships with autistic people and greater interest in hanging out with or starting a conversation with autistic adults evaluated in first impression videos. These higher levels of expressed social interest in autistic adults occurred for NA participants completing autism acceptance training despite no improvements on four of the six trait assessments, suggesting that the training largely did not alter negative evaluations of autistic social presentation differences, but may have lessened the salience or importance of these evaluations for predicting social interest.

Collectively, the effects reported here for autism acceptance training are consistent with previous studies indicating that exposure to realistic representations of people with disabilities can reduce misconceptions about them (Gillespie-Lynch et al., 2015; Moore & Nettelbeck, 2013; Schwartz et al., 2010). Furthermore, although the broader social psychology literature indicates that meaningful

Table 5. Correlations between explicit and implicit measures of autism stigma.

	Autism knowledge	Autism stigma	SLOF Interpersonal	SLOF Acceptability	SLOF Activities	SLOF Work	IAT D-score
Autism knowledge[a]	1	–0.41**	0.21**	0.31**	0.39**	0.38**	0.02
Autism stigma[b]	–	1	–0.17*	–0.27**	–0.32**	–0.38**	–0.04
SLOF Interpersonal	–	–	1	0.24**	0.43**	0.43**	–0.11
SLOF Acceptability	–	–	–	1	0.37**	0.43**	–0.12
SLOF Activities	–	–	–	–	1	0.71**	–0.11
SLOF Work	–	–	–	–	–	1	–0.1
IAT D-score	–	–	–	–	–	–	1

SLOF: Specific Levels of Functioning Scale; IAT: Implicit Association Test.
[a]Scores derived from the Autism Knowledge Scale (Gillespie-Lynch et al., 2015).
[b]Scores derived from the Social Distance Scale (Gillespie-Lynch et al., 2015).
*$p < 0.05$.
**$p < 0.01$.

interpersonal contact with individuals from marginalized groups is the most effective mechanism for reducing prejudice and improving attitudes (Dasgupta & Rivera, 2008; Jackson et al., 2014; Pettigrew & Tropp, 2008), the current training program raises the possibility that even indirect exposure to autism and autistic people may be able to produce measurable benefits for counteracting common stereotypes and combating exclusionary attitudes, at least within laboratory settings. Importantly, given that NA adults completing the general mental health training did not differ meaningfully in stigma and negative attitudes about autism relative to those in a no-training control condition, simply increasing exposure and sensitivity to clinical conditions more generally appears insufficient for combating autism-related stigma. Rather, including factual knowledge specifically about autism and presenting engaging first-person accounts from autistic people detailing their strengths and challenges may be necessary for a training program to improve attitudes about autism.

Previous studies have found that autism knowledge among NA adults is responsive to training (Gillespie-Lynch et al., 2015) and is related to less stigma about autism (Gillespie-Lynch et al., 2015) and more favorable assessments of autistic people (Sasson & Morrison, 2019). The current findings extend on this prior work by demonstrating that a brief training module focused on autism acceptance results not only in higher autism knowledge among NA adults but also lower explicit biases about autism and more positive first impressions of real autistic people compared to a general mental health training module and a no-training control condition. Specifically, biases were lower for judgments commonly associated with both interpersonal (e.g. attractiveness, social interest, and norms of acceptable behavior) and professional (e.g. intelligence, functional abilities, and work skills) discrimination of autistic people. Employers often express negative beliefs about the work abilities of autistic adults, including concerns about productivity (Mai, 2019) and dependability (Nicholas et al., 2019), and previous work has shown that contact with autistic people can help lessen common stereotypes employers often hold about autistic people and increase their confidence in their abilities (Nicholas et al., 2019). This suggests that exposing NA employers and coworkers to realistic representations of autistic people, such as those included in the autism acceptance training used here, may contribute to an increased understanding of autism in the workplace. Such broader personal and professional benefits are currently speculative. Further research is needed to determine whether the benefits of training reported here extend beyond the confines of experimental testing and translate to real-world improvements for autistic people.

Importantly, participants completing the autism acceptance training exhibited lower explicit biases in their first impressions of autistic adults regardless of whether a diagnostic label was provided. This suggests that the effects of the training were not constrained to participants being primed by the presence of a diagnostic label to "check" and suppress their biases, but rather increased their sensitivity and decreased their negative inferences about autistic characteristics regardless of whether they were informed that the person was autistic. Alternatively, it is possible that the autism acceptance training primed them to assume everyone they evaluated was autistic whether or not a label was included. Interestingly, including a diagnostic label resulted in more favorable first impressions across all training conditions, including the autism training condition. This is consistent with prior research (Brosnan & Mills, 2016; Morrison et al., 2019; Sasson & Morrison, 2019) and may occur because diagnostic awareness provides an explanatory context for behaviors or mannerisms perceived as non-normative.

Despite the autism acceptance training resulting in lower explicit biases about autism relative to the general mental health training condition and the no-training control, it exerted no effect on implicit biases. Across all

groups, NA participants showed significantly faster response times when autism-related labels were paired with negative personal attributes compared to when they were paired with positive personal attributes, in line with previous findings of implicit biases toward autistic individuals (Dickter, Burk, Zeman, et al., 2020). In contrast to explicit biases that are consciously held, evolve quickly through learning or personal experiences, and are constrained by social desirability, implicit biases reflect underlying beliefs and automatic associations formed over time through the reinforcement of stereotypes, which tend to be more resistant to change (Devine et al., 1989). Previous work on racial biases has found that while the *quality* of contact with outgroup members is important for improving explicit biases, improvements in implicit biases depend instead on the *quantity* of contact (Dasgupta & Rivera, 2008; Gardiner & Iarocci, 2014; Prestwich et al., 2008), suggesting that short-term training programs may not provide the level of contact needed to elicit changes in implicit biases.

Indeed, a wealth of evidence has accumulated highlighting the inefficacy of training programs to reduce implicit biases despite their widespread adoption (Fitzgerald et al., 2019), with only protracted, multi-faceted, and "habit-breaking" interventions producing sustained benefits (Devine et al., 2012). Future research is needed to determine if such an approach may lead to reductions in implicit biases about autism. Work examining the impact of implicit biases about autism is limited, but related research suggests that implicit biases about mental health diagnoses can influence diagnostic and treatment-related decisions made by trained clinicians (Peris et al., 2008; Stull et al., 2013). How implicit biases may affect the social, professional, and clinical experiences of autistic people remains an understudied area worthy of continued examination. Additional research is also needed to determine whether reducing explicit biases without also reducing implicit biases among NA individuals still confers benefit for the lived experiences of autistic people.

These findings should be interpreted in the context of several limitations. First, because attitudes toward autism were assessed using questionnaires and brief videos of autistic people, it is unknown whether the benefits of autism acceptance training would persist over time and translate to real-world improvements for interactions between autistic and NA adults. Furthermore, even if decreased explicit biases did translate to more inclusive attitudes toward autistic people in the real world, they may not produce noticeable benefits for the lives of autistic people. Future work is encouraged to assess whether the training effects reported here extend beyond a laboratory setting, are maintained over time, and produce appreciable benefits for autistic people. Second, the videos of autistic adults used in the study predominantly consisted of White males with IQs in the average to above-average range. Our findings may have differed if a more diverse sample of autistic people was included. Given prior evidence suggesting that implicit racial biases may contribute to differences in how observers perceive autistic traits in Black and White children (Obeid et al., 2020), future work examining the intersection between racial and diagnostic biases in autism is encouraged. Finally, recruitment of NA participants was restricted to undergraduate university students currently enrolled in psychology or neuroscience courses. Although a significant strength in one respect—findings may highlight the potential benefit of implementing training programs in university settings for improving inclusivity of autistic students—college students may also differ in important ways from NA individuals in the broader population. They may have greater autism familiarity and knowledge, express more inclusive attitudes, and be more accepting of differences relative to older adults, which may limit generalizability of the results reported here. Higher a priori knowledge about autism may have also contributed to the lack of a significant difference in overall autism knowledge between the autism acceptance training and the no-training control conditions ($p = 0.07$). However, effects were found at the item level, with participants who completed the training being significantly less likely to endorse several misconceptions about autism than the control group. Autism knowledge was also significantly higher among participants completing autism acceptance training relative to general mental health training, further validating the benefit of targeted training.

Despite these limitations, this study highlights the potential of a brief video-based training module for reducing explicit biases about autism among NA adults. Whereas a general mental health training module produced few benefits to autism understanding relative to a no-training control, NA participants completing an autism acceptance training expressed lower explicit biases and stigma toward autism and higher social interest in interacting with autistic adults. However, the presence or magnitude of implicit biases related to autism was unaffected by autism acceptance training, with implicit associations between autism-related labels and negative personal attributes persisting for participants across all training conditions. These findings suggest that autism acceptance training offers significant promise for promoting more inclusive attitudes toward autistic people among NA adults, but these benefits may be limited to consciously controlled responses. Future studies are needed to determine whether benefits are sustained over time and extend to real-world environments and interactions.

Acknowledgements

We thank Dr Grace Iarocci and Dr Dominic Trevisan for providing the autism training module that was modified for this study, and Evan "Silky" Cosentino for providing the narration for the

autism acceptance training video. We would also like to thank Dr Robert Ackerman for his assistance with statistical analysis, as well as Alejandra Castillo, Danny Dunn, Jay De La Garza, Kathryn King, Isabel Magaña, Hulon Sherard, Priya Vanparia, Nick Via, and Mattie Watts for their assistance with data collection and input on study design and interpretations.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This research was funded by the Texas Higher Education Coordinating Board's Autism Grant Program.

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